Biomechanics in basketball

Time 4
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I. Brief Description of Selected Movement

The selected movement is a basketball shooting action analyzed around the ball release phase. For each frame, we assume the ball could be released and follow a projectile trajectory defined by its position, velocity, and angle.

For every possible release frame, the minimum required speed to reach the basket is calculated, representing the optimal release velocity since it minimizes energy use and improves control. The player's actual release velocity is then compared to the optimal value to measure performance error.

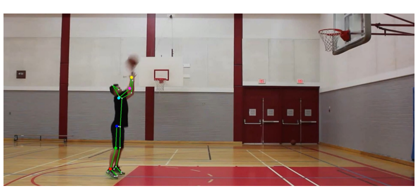
The analysis also considers the relationship between release height and optimal angle, where higher release points generally require a slightly smaller angle.

The jump contributes by increasing the release height, which typically reduces the required optimal velocity and slightly lowers the optimal release angle.

II. Methodology

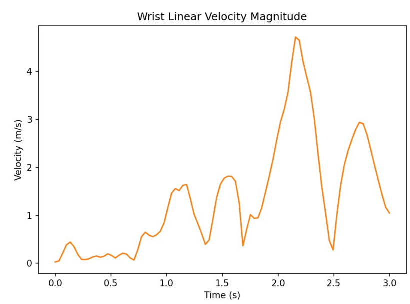
The motion analysis was conducted using Sports2D integrated with Python for kinematic data extraction and post-processing. Video calibration was performed using the built-in Sports2D scaling method by entering the athlete's known height to convert pixel coordinates into real-world units. Body joint trajectories were obtained through markerless pose estimation, while the basketball trajectory was tracked using computer vision techniques to extract its centroid position frame-by-frame. Linear and angular kinematic variables were then computed from the calibrated positional data for subsequent projectile and release optimization analysis.

III. Results

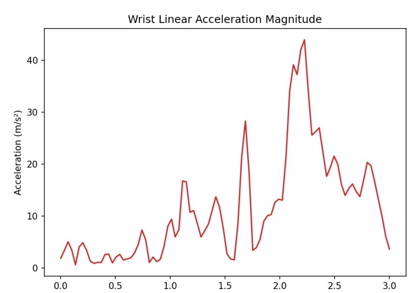


(a) A basketball player is captured mid jump during a shot

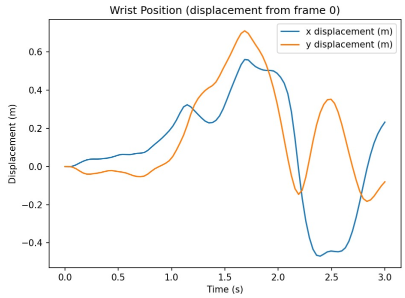
(a) A skeletal overlay mapping their joint positions.



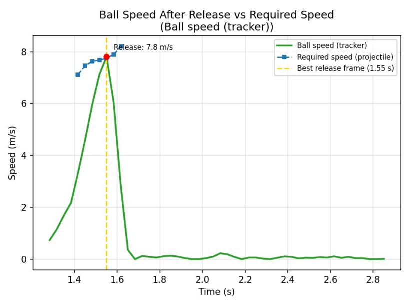
(a) The graph shows wrist velocity over time, fluctuating to a peak at ball release



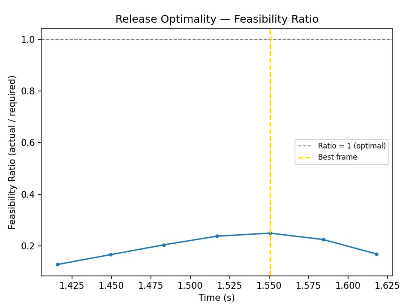
(a) The graph shows wrist acceleration over time, peaking sharply just before the ball is released.



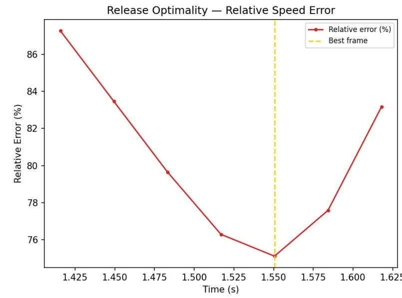
(a) The graph illustrates the Wrist Position (displacement from frame 0) over time, tracking both the x and y displacement in meters.



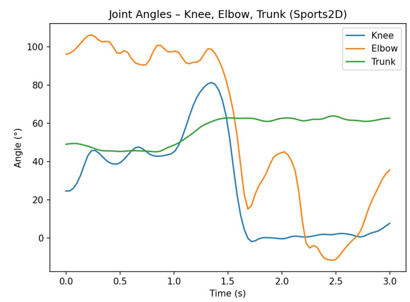
(a) This graph compares the measured speed of a basketball after release against the theoretically required speed for a successful shot, highlighting the optimal release frame.



(a) This graph illustrates the feasibility ratio (actual speed divided by required speed) over time, identifying the moment when the ball's speed is closest to the required speed for a successful shot.



(a) This graph illustrates the relative speed error over time, showing a minimum value at the optimal release frame.



(a) This graph illustrates the angles of the knee, elbow, and trunk over time during a basketball shooting motion.

IV. Kinematic Interpretation [3]

A. *Kinematic Analysis:*

1) ***Angular Kinematics (The Shot):*** The "wrist flick" is analyzed using angular acceleration. We will measure the rate of change of the wrist's angular velocity to find the point of maximum acceleration.

2) ***General Motion:*** Basketball shot is a combination of linear and angular motion, which is typical for human movement (translation of the whole body via the jump and rotation of the joints via the shot).

B. *Projectile Motion and Optimal Performance*

1) ***Projectile Laws:*** Once the ball is released, it becomes a projectile governed by the laws of gravity, moving in a parabolic path.

2) ***Optimization Variables:*** The calculation of minimum possible speed involves calculating projection velocity (V.) and projection angle (θ) to achieve the required range (R) and maximum height (H) to clear the hoop.

C. *Motion description*

1) ***Global (Inertial) Reference Frame:*** Use a fixed X-Y axis to track the trajectory of the ball and the linear height of the player.

2) ***Relative Reference System:*** Use this to describe the movement of the hand relative to the forearm. This is essential for calculating the specific joint angle and angular acceleration of the wrist flick.

D. *Frame-Based Optimal Release Analysis*

1. Externally (Osteokinematics)

This analysis models the visible motion of the player's joints during shooting by assuming that, at each frame, the ball could be released and follow a projectile trajectory based on its instantaneous position, velocity, and angle. For every possible release moment, the minimum required release speed to reach the basket is calculated, representing the optimal velocity since it minimizes kinetic energy.

$$KE = \frac{1}{2}mv^2 \tag{1}$$

The player's actual release velocity is then compared to this optimal value using percentage error to quantify deviation from optimal performance. The model also accounts for release height, showing that higher release points generally correspond to slightly smaller optimal angles and reduced required speed.

$$\text{Error }(\%) = \frac{|v_{\text{actual}} - v_{\text{optimal}}|}{v_{\text{optimal}}} \times 100 \tag{2}$$

2. Internally (Arthrokinematics)

This describes the internal joint mechanics occurring during the shot.

Rolling & Gliding: As the wrist moves during ball release, the carpal bones roll and glide smoothly over each other. Proper arthrokinematic motion ensures efficient force transmission from the forearm to the hand, contributing to controlled ball release.

V. Scientific Support

A. *Research Claim*

This study claims that there exists an optimal launch velocity that minimizes the mechanical energy required for a basketball shot while still allowing the ball to successfully reach the basket. Since kinetic energy is proportional to the square of velocity,

$$KE = \frac{1}{2}mv^2,$$

reducing the required launch velocity results in a more energy-efficient shot. In real basketball conditions, the optimal release angle is not necessarily 45°, as assumed in ideal projectile motion, because the release height differs from the rim height. Instead, the optimal angle depends on shooting distance and release height and is typically greater than 45°. Therefore, the minimum required launch velocity is a function of shooting distance, release height, and the corresponding release angle that minimizes speed under these constraints.

B. *Literature Review*

*1) **Optimal Angle in Basketball Free Throw**:* The study [1] analyzes the optimal release angle and initial velocity for a successful basketball free throw using projectile motion and Newtonian mechanics. By applying optimization techniques and anthropometric data from professional players, it determines the release angle that minimizes the required launch velocity. The findings show that the optimal shot does not necessarily occur at 45°, but typically falls between 40° and 55°, depending on release height, shooting distance, and biomechanical factors, with the best shot being the one that requires the minimum launch velocity.

$$\theta_{\text{opt}} = \tan^{-1}\left(\frac{y_b - y_0}{x}\right) + 45^\circ$$

*2) **Most Energy-Efficient Basketball Shot**:* The study [2] presents a mathematical model for determining the most energy-efficient basketball shot from any court position. Using kinematics and calculus-based optimization, it identifies the minimum launch velocity and corresponding optimal release angle for a successful shot. The results show that efficiency is achieved by minimizing launch velocity, that the optimal angle decreases as shooting distance increases, and that required velocity increases approximately linearly with distance. Despite simplifying assumptions, the study provides a general theoretical framework for analyzing basketball shooting mechanics.

C. *Review of Theoretical Mechanics Analysis of Basketball Shooting*

The paper [4] develops a physics-based model of basketball shooting using projectile motion equations. It relates release velocity, release angle, release height, and shooting distance under gravitational acceleration. The study shows that the optimal release angle is typically greater than 45° due to the difference between release and rim height, and confirms the existence of a minimum required release velocity for a successful shot. Since kinetic energy is proportional to v^2, minimizing release velocity reduces the mechanical energy required. Overall, the paper provides a theoretical foundation for energy-efficient basketball shooting.

$$KE = \frac{1}{2}mv^2,$$

D. *Connection to This Study*

Previous theoretical and biomechanical studies support modeling basketball shooting as a velocity-minimization (energy-efficient) problem. Research shows that a minimum release velocity exists for a successful shot, with the optimal release angle typically greater than 45° due to elevated release height. Biomechanical findings further indicate that greater release height, proper joint coordination, and timing at peak jump reduce the required velocity, improving efficiency and control.

References

[1] M. Author, "Analysing an Optimal Angle in Basketball Free Throw," unpublished manuscript, 2024.

[2] M. Author, "General Form Equation for the Most Energy-Efficient Basketball Shot," unpublished manuscript, 2024.

[3] R. Caseiro, "Kinematic analysis of the basketball jump shot with increasing shooting distance," 2023. [Online]. Available: https://doi.org/XXXX

[4] A. Author, "A theoretical mechanics analysis of shooting basketball,"